

Cooperative Decision-Making for Multi-UUV Target Tracking to Ensure Persistent Detection Coverage

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Abstract—Persistent tracking of dynamic targets using a fleet of Unmanned Underwater Vehicles (UUVs) remains a challenging problem, primarily due to the dynamic nature of the environment and the constraints of underwater communication. This paper addresses the challenge of cooperative decision-making for multi-UUV systems to ensure the target remains within a predefined detection range for as long as possible. We formulate this problem as a global optimization problem where the objective is to maximize the cumulative tracking time. Unlike traditional local optimization methods, we propose a real-time decision-making strategy based on a Genetic Algorithm (GA). The GA optimizes the heading and speed commands for each UUV by evaluating potential future trajectories against the predicted target movement. Simulation results demonstrate that the proposed GA-based method significantly outperforms a benchmark greedy strategy, achieving a longer tracking duration and more robust performance against evasive target maneuvers.

Keywords—Multi-UUV systems, cooperative target tracking, Genetic Algorithm

I. INTRODUCTION

Unmanned Underwater Vehicles (UUVs) have become indispensable tools for a wide range of underwater missions, from oceanographic exploration to defense and security applications. A critical capability within this domain is cooperative target tracking, where a fleet of UUVs collaboratively monitors a moving target to ensure persistent detection coverage [1, 2]. This task is particularly challenging due to the complex and dynamic nature of the underwater environment, which is characterized by uncertain ocean currents, limited and unreliable acoustic communication, and the inherent constraints of UUV platforms such as limited energy and sensor range [3, 4].

The core of multi-UUV cooperative tracking lies in solving a complex decision-making problem that intertwines state estimation and motion planning. The system must accurately estimate the target's state (position, velocity) and then plan optimal trajectories for the UUVs to maintain that target within their collective sensor footprint. Research in this field has evolved along several parallel tracks, primarily focusing on decentralized control architectures, advanced estimation algorithms, and intelligent planning strategies to overcome the aforementioned challenges. In terms of perception and state estimation, researchers have developed sophisticated algorithms to handle the "bearing-only" measurements and highly-maneuvering targets common in underwater scenarios. For instance, distributed fusion algorithms based on the Extended Kalman Filter (EKF) and Interacting Multiple Model (IMM) frameworks have been employed to fuse data from multiple UUVs, yielding more accurate and robust target state estimates [5]. To further combat error accumulation in anchor-free environments, some

studies have innovatively integrated current measurements as external references, significantly improving the accuracy of both self-localization and target tracking [6, 8]. For the decision-making and path planning layer, the research community has explored a spectrum of methods. Early approaches often relied on optimization-based algorithms. For example, hybrid algorithms combining Improved Multi-Objective Particle Swarm Optimization (IMOPSO) with a Dynamic Window Approach (DWA) have been proposed for efficient 3D path planning in complex terrains and currents [1]. Similarly, Particle Swarm Optimization (PSO) has been used for online collaborative optimization of tracking paths based on predicted target states [4].

More recently, machine learning, particularly Multi-Agent Reinforcement Learning (MARL), has emerged as a powerful paradigm for enabling adaptive and intelligent cooperative behaviors. Several studies have constructed collaborative task decision-making models based on algorithms like MATD3 (Multi-Agent Twin Delayed Deep Deterministic Policy Gradient), which allow UUVs to possess autonomous decision-making capabilities even under weak communication conditions [2, 10]. To address the scalability issues of MARL, innovative architectures like Elastic Software-Defined MARL (ESD-MARL) and Incremental MARL based on Minimal Reward Participation (IMARL-MRP) have been introduced, enabling AUV clusters to expand arbitrarily without retraining [10]. Other learning-based approaches include Autonomous Cooperative Search Learning Algorithms (ACSLA) [6] and integrated algorithms combining bio-inspired neural networks with tracking control [7].

Notably, Genetic Algorithms (GAs) and other evolutionary computation methods have demonstrated significant potential in solving the complex optimization problems inherent in path planning. While extensively applied to UAVs for generating optimal paths in obstacle-rich environments [13, 14, 15, 16], their principles are highly relevant to UUV operations. Research has directly explored GAs for multi-target path planning of UUVs [17] and even hybridized GA with Dynamic Programming (DP) to develop planners that outperform standard GA in both speed and solution quality [18]. Despite these advancements, achieving robust, real-time, and scalable cooperative tracking in the face of severe environmental uncertainties and communication constraints remains an open challenge. Many existing algorithms face a trade-off between computational complexity and performance, especially when scaling to larger teams of heterogeneous UUVs.

Motivated by this landscape, this paper investigates a decentralized cooperative decision-making framework for multi-UUV target tracking. Building upon the proven global search capabilities of genetic algorithms in path planning [13,

14, 17, 18], our core contribution is a GA-based optimization module. This module integrates real-time target kinematics (position, velocity, direction) and the state of the UUV cluster (positions, headings, speed limits) to compute optimal control commands for each UUV. The primary objective is to dynamically maximize the cumulative time the target remains within a predefined effective detection range (1 kilometer), thereby ensuring persistent detection coverage.

The remainder of this paper is organized as follows: Section 2 formally defines the problem and presents the system models. Section 3 details the proposed GA-based cooperative decision-making algorithm. Section 4 presents and discusses the simulation results, and Section 5 concludes the work.

II. SYSTEM MODELING FOR UUV TRACKING

This section details the mathematical models governing the multi-UUV cooperative target tracking system. The primary objective is to formalize the problem of maximizing the cumulative tracking time by controlling the headings and speeds of a UUV fleet, subject to kinematic, sensing, and communication constraints.

A. System Overview

The considered scenario involves a fleet of N homogeneous UUVs tasked with cooperatively tracking a single dynamic target. Each UUV can control its forward speed and heading. The UUVs can communicate their own states and perceived target information with neighbors within a limited communication range to facilitate cooperative decision-making. The tracking mission is considered successful at any given time if the target lies within the detection radius of at least one UUV.

B. UUV Kinematic Model

Given the focus on strategic decision-making over low-level dynamics, a discrete-time kinematic model is adopted for each UUV. The state of the i -th UUV at time step k is defined by its position $(x_i(k), y_i(k))$ and its velocity vector, characterized by speed $v_i(k)$ and heading angle $\theta_i(k)$.

The motion of UUV i is governed by:

$$\begin{cases} x_i(k+1) = x_i(k) + v_i(k) \cdot \cos(\theta_i(k)) \cdot \Delta T \\ y_i(k+1) = y_i(k) + v_i(k) \cdot \sin(\theta_i(k)) \cdot \Delta T \end{cases} \quad (1)$$

Where ΔT is the sampling time interval. The control inputs for UUV i are its speed change $\Delta v_i(k)$ and heading change $\Delta \theta_i(k)$, leading to:

$$\begin{cases} v_i(k+1) = v_i(k) + \Delta v_i(k) \\ \theta_i(k+1) = \theta_i(k) + \Delta \theta_i(k) \end{cases} \quad (2)$$

These control inputs are constrained by the UUVs' physical limitations:

$$\begin{cases} v_{\min} \leq v_i(k) \leq v_{\max}, \quad \forall i, k \\ |\Delta v_i(k)| \leq \Delta v_{\max}, \quad \forall i, k \\ |\Delta \theta_i(k)| \leq \Delta \theta_{\max}, \quad \forall i, k \end{cases} \quad (3)$$

C. Target Motion Model

The target's state at time k is denoted as $\mathbf{p}_t(k) = (x_t(k), y_t(k))$. Its motion is assumed to be unknown

but predictable over a short horizon for planning purposes. A common model for unpredictable targets is a constant-velocity model with random perturbations, which can be used for prediction within the optimization horizon H :

$$\mathbf{p}_t(k+1) = \mathbf{p}_t(k) + \mathbf{v}_t(k)\Delta T + \mathbf{w}(k) \quad (4)$$

where $\mathbf{v}_t(k)$ is the estimated target velocity and $\mathbf{w}(k)$ represents process noise, accounting for target maneuvers and uncertainties.

D. Sensing and Communication Models

Each UUV is equipped with a sensor (e.g., sonar) with a limited detection range R_d . The binary detection function $D_i(k)$ for UUV i and the collective coverage function $C(k)$ for the fleet are defined as:

$$D_i(k) = \begin{cases} 1 & \text{if } \|\mathbf{PU}_i(k) - \mathbf{PT}(k)\| \leq R_d \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

$$C(k) = \begin{cases} 1 & \text{if } \sum_{i=1}^N D_i(k) \geq 1 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where $\mathbf{PU}_i(k)$ denotes the position of the UUV i at time k , and $\mathbf{PT}(k)$ represents the position of the target at time k . Successful tracking at time k is achieved when $C(k) = 1$.

UUVs can share data (e.g., own state, target estimates) with others within a communication range R_c . The communication topology is represented by a time-varying graph $G(k) = (V, E(k))$, where an edge $(i, j) \in E(k)$ exists if $\|\mathbf{p}_i(k) - \mathbf{p}_j(k)\| \leq R_c$. This work assumes that essential cooperative data can be exchanged reliably within this network.

E. Collaborative Control Objective

The core objective is to maximize the total time the target is under surveillance by the UUV fleet over a mission duration of K steps. This is formalized as a constrained optimization problem.

Let $\mathbf{U}(k) = [\Delta v_1(k), \Delta \theta_1(k), \dots, \Delta v_N(k), \Delta \theta_N(k)]^T$ be the collective control vector for all UUVs at time k . The goal is to find the sequence of control inputs $\{\mathbf{U}(k)\}_{k=0}^{K-1}$ that maximize the follow objective:

$$f_1 = \sum_{k=1}^K C(k) \quad (7)$$

When UUV i is located far from the target—specifically, when the target lies outside the UUV's detection range—it can only acquire the target's position through inter-vehicle communication. In such scenarios, the UUV must be guided toward the target to enable successful tracking at a future time step. To this end, a secondary objective function f_2 is introduced to minimize the distance between UUV i and the target. The expression for f_2 is given as follows:

$$f_2 = \|\mathbf{PU}_i(k) - \mathbf{PT}(k)\| \quad (8)$$

Since the evaluation of the objective function f_1 inherently depends on the distances between the target and the UUVs, the effectiveness of the UUVs' cooperative strategy can be assessed using f_2 alone. Over a prediction horizon of

H future time steps, the overall objective function J for cooperative target tracking is formulated as follows:

$$\begin{aligned} \min \quad & J = \sum_{k=1}^H \sum_{i=1}^N \|\mathbf{PU}_i(t+k) - \mathbf{PT}(t+k)\| \\ & v_{\min} \leq v_i(k) \leq v_{\max}, \quad \forall i, k \\ & |\Delta v_i(k)| \leq \Delta v_{\max}, \quad \forall i, k \\ & |\Delta \theta_i(k)| \leq \Delta \theta_{\max}, \quad \forall i, k \\ & \|\mathbf{PU}_i(k) - \mathbf{PU}_j(k)\| \leq R_c \end{aligned} \quad (9)$$

The constraints ensure that the UUV's speed and heading rate of change adhere to realistic physical limitations, and the communication graph $G(k)$ must remain connected—or, at a minimum, no UUV should be isolated from the main group.

This optimization problem is high-dimensional, nonlinear, and non-convex, making it suitable for a global optimization method like the Genetic Algorithm proposed in this paper. The GA will operate in a receding-horizon fashion, solving for optimal control sequences over a predictive horizon H at each decision point.

III. COOPERATIVE DECISION-MAKING METHOD BASED ON GENETIC ALGORITHM

This section presents the detailed implementation of the Genetic Algorithm (GA) for solving the multi-UUV cooperative tracking optimization problem formulated in Section 2. The GA is particularly suitable for this high-dimensional, nonlinear, and non-convex optimization problem due to its powerful global search capabilities.

A. Algorithm Framework

The proposed GA operates within a receding horizon control framework, as illustrated in Figure 1. At each decision interval, the algorithm utilizes the current states of all UUVs and the target to compute optimal control sequences over a finite prediction horizon H . Only the first control action from the optimized sequence is executed by each UUV. This process repeats at subsequent time steps with updated state information, enabling adaptive and real-time decision-making in dynamic environments.

B. Chromosome Encoding

A chromosome represents a potential solution to the optimization problem, encoding the collective control strategy for the entire UUV fleet over the prediction horizon H . Each chromosome is implemented as a real-valued vector.

For a system with N UUVs and a prediction horizon of H steps, the chromosome *Chrom* is structured as:

$$\begin{aligned} \text{Chrom} = & [v_1(1), \theta_1(1), v_2(1), \theta_2(1), \dots, v_N(1), \theta_N(1), \dots \\ & v_1(H), \theta_1(H), v_2(H), \theta_2(H), \dots, v_N(H), \theta_N(H)] \end{aligned} \quad (10)$$

The total length of the chromosome is $2 \times N \times H$, directly representing all decision variables in the optimization problem.

C. Fitness Function Design

The fitness function evaluates the quality of each chromosome based on the objective function defined in Equation (9), with additional penalty terms for constraint violations.

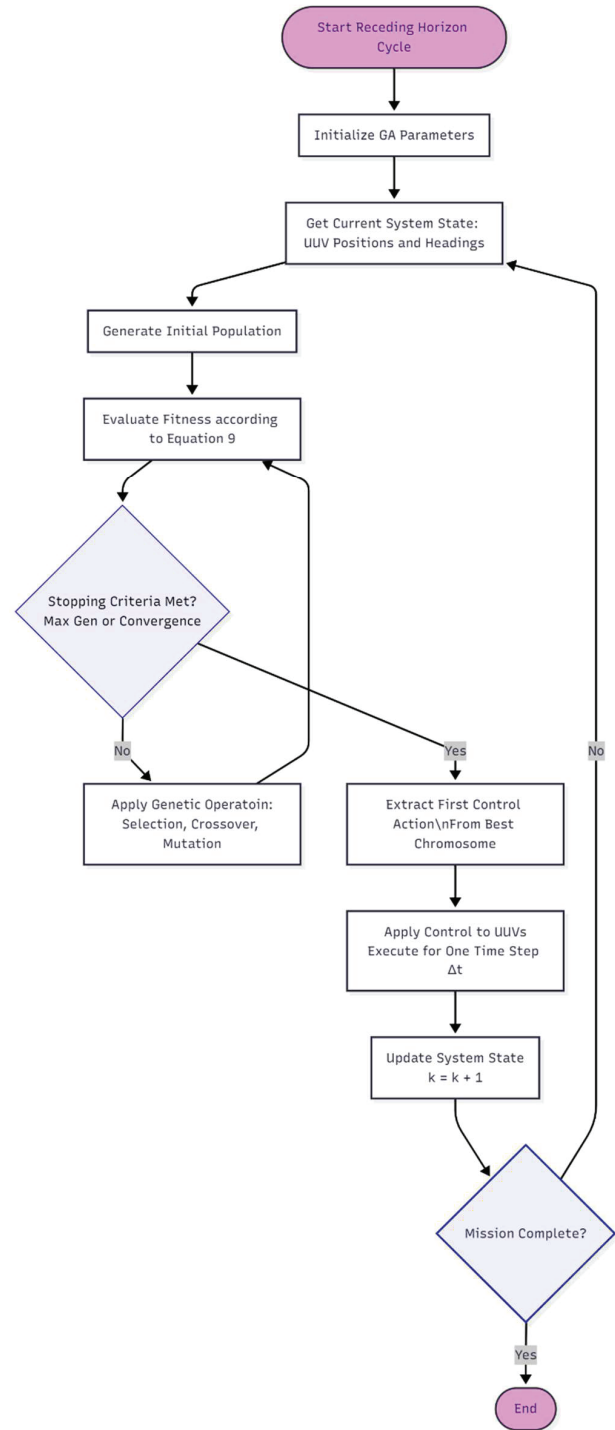


Figure 1. Flowchart of the GA-based receding horizon control framework

• Primary Objective Component

The core fitness component directly implements the minimization objective from Equation (9):

$$F_{\text{primary}} = \sum_{k=1}^H \sum_{i=1}^N \|\mathbf{p}_i(k) - \mathbf{p}_t(k)\| \quad (11)$$

Where $\mathbf{p}_i(k)$ and $\mathbf{p}_t(k)$ represent the positions of UUV i and the target at future time step k , predicted using the control sequence defined by the chromosome.

- **Constraint Penalty Components**

To ensure feasible solutions, control input bounds penalty and communication connectivity penalty are incorporated for constraint violations:

$$P_{bounds} = \sum_{k=1}^H \sum_{i=1}^N \left[\max \left(0, \frac{|\Delta v_i(k) - \Delta v_{\max}|}{\Delta v_{\max}} \right)^2 + \max \left(0, \frac{|\Delta \theta_i(k) - \Delta \theta_{\max}|}{\Delta \theta_{\max}} \right)^2 \right] \quad (12)$$

$$P_{comm} = \sum_{k=1}^H \sum_{i=1}^{N-1} \sum_{j=i+1}^N \max \left(0, \frac{\|\mathbf{p}_i(k) - \mathbf{p}_j(k)\| - R_c}{R_c} \right)^2 \quad (13)$$

- **Complete Fitness Function**

The final fitness function combines all components:

$$Fitness(Chrom) = F_{primary} + \lambda_b P_{bounds} + \lambda_c P_{comm} \quad (14)$$

Where λ_b and λ_c are weighting coefficients that balance objective performance with constraint satisfaction.

IV. SIMULATION EXPERIMENTS AND RESULT ANALYSIS

To validate the effectiveness of the proposed cooperative decision-making method based on the Genetic Algorithm (GA) for multi-UUV target tracking, we conducted simulation experiments using MATLAB/Simulink environment. The simulations involve a fleet of 24 UUVs cooperatively tracking a single dynamic underwater target in a 2D plane (assuming depth is constant for simplicity). The experiments focus on persistent detection coverage, fleet connectivity maintenance, and adaptation to target maneuvers under realistic physical constraints.

A. Simulation Setup

The mission was conducted within a $180\text{km} \times 180\text{km}$ underwater operational area, and the total simulation duration was set to 5000 seconds, utilizing a discrete time step of $\Delta T=5$ second for system state updates and control computation. The UUV and Target Parameters are listed in Table 1.

TABLE I. UUV AND TARGET PARAMETERS

	$v_{\max}$ (m/s)	$\Delta v_{\max}$ (m/s ²)	$\Delta \theta_{\max}$ (rad/s)	R_c (m)	R_d (m)
UUV	16	0.3	0.26	5000	1000
Target	25	0.1	0.05		

During the initialization phase of the Genetic Algorithm, key parameters were configured as follows: the population size was set to 200 individuals to maintain sufficient genetic diversity while preserving computational efficiency; the maximum number of generations was fixed at 100 to ensure convergence within a practical time frame; the crossover and mutation probabilities were set to 0.8 and 0.1, respectively, to strike an effective balance between exploration and exploitation of the search space; and the prediction horizon was defined as 20 time steps to provide adequate foresight within the receding horizon control framework. Furthermore,

penalty weights for constraint handling were initialized with $\lambda_b = 10$ for boundary constraints and $\lambda_c = 5$ for communication connectivity constraints, ensuring appropriate scaling of constraint violations during fitness evaluation. This parameter configuration forms the foundation for the algorithm's evolutionary optimization process in addressing the multi-UUV cooperative tracking problem.

B. Simulation Results and Analysis

This paper first presents a simulation of the dynamic tracking behavior of individual unmanned underwater vehicles (UUVs) as a target traverses a linear formation of the UUV cluster, with the results illustrated in Fig. 2. Owing to the target's significantly higher speed and directional velocity compared to those of the UUVs, it remains within the tracking range of the cluster only transiently before rapidly exiting. The simulation results demonstrate that, in practical deployment scenarios, arranging UUVs in a single-file linear configuration should be avoided, as such an arrangement lacks the spatial coverage and responsiveness required to effectively track fast-moving targets.

As the target enters the dispersed UUV cluster from the lower-left region, the tracking process by the UUVs is illustrated in Figs. 3–5. In these figures, red-colored UUVs indicate those that have entered the target's tracking range. The simulation results demonstrate that the UUV cluster maintains continuous tracking of the target over a distance of 40 kilometers.

When the target enters the spatially dispersed UUV cluster from the upper-left direction, the cooperative tracking process is illustrated in Figs. 6–10. The simulation results indicate that, in order to maximize tracking duration, each UUV navigates toward the predicted direction of the target's motion. Under this coordination strategy, the UUV cluster achieves a sustained tracking range of approximately 50 kilometers.

V. CONCLUSION

The proposed GA-based cooperative decision-making strategy enables a multi-UUV fleet to persistently track a dynamic target. Simulations confirm the method significantly extends tracking duration, achieving up to 50 km of continuous coverage by dynamically optimizing UUV control within a receding horizon framework. It outperforms static formations and effectively enforces physical and communication constraints, proving its robustness, real-world applicability, and superior autonomous collaborative tracking capability.

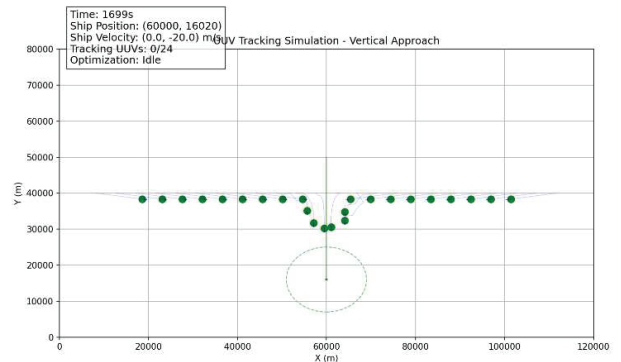


Fig 2. Target traversing through the UUV cluster: linear-formation tracking results

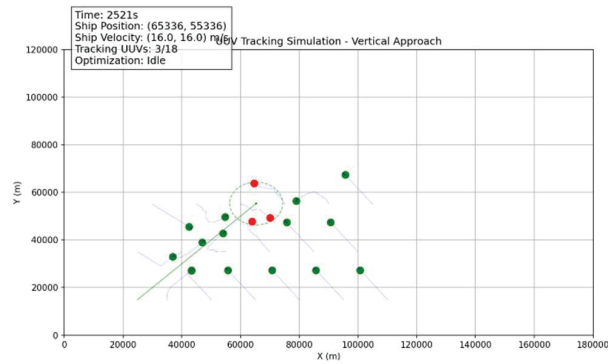


Fig 3. Target entry from lower-left: initial tracking phase

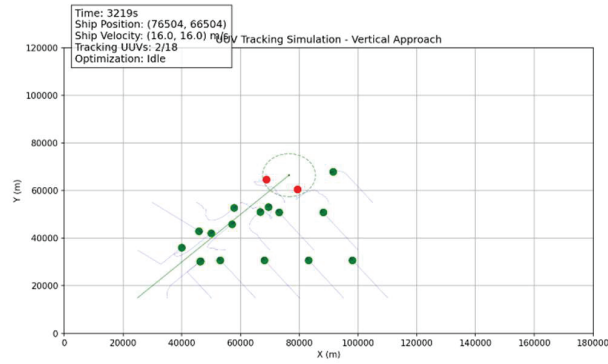


Fig 4. Target tracking in progress: cooperative engagement

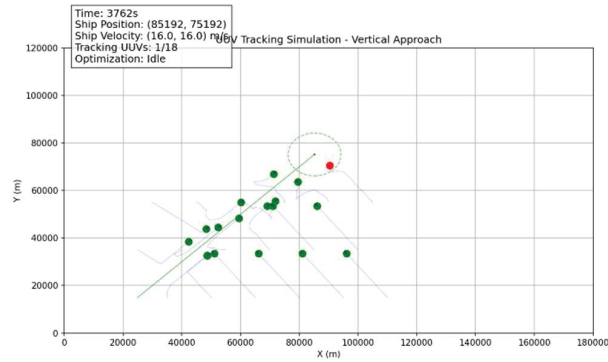


Fig 5. Sustained tracking achievement: 40 km coverage range

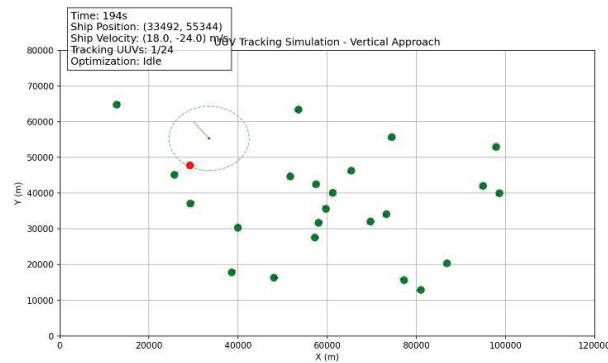


Fig 6. Target entry from upper-left: predictive repositioning

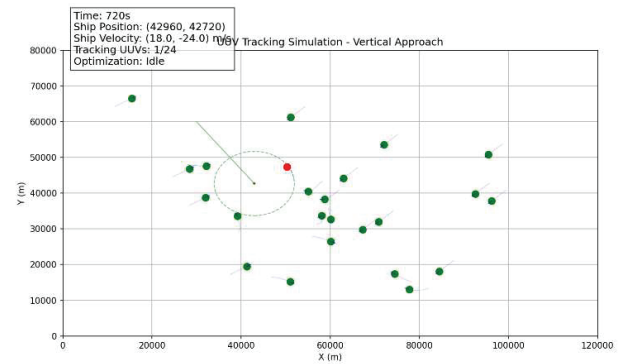


Fig 7. Cluster coordination: converging on predicted path

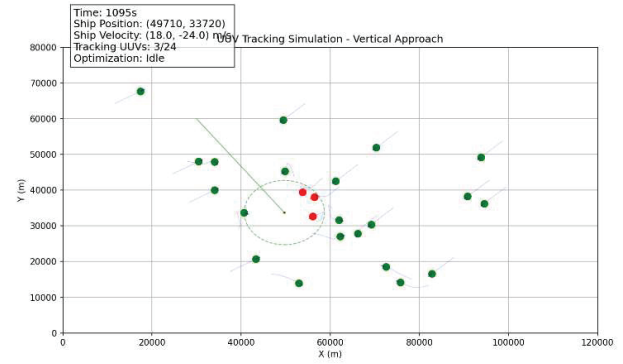


Fig 8. Adaptive formation: maintaining tracking coverage

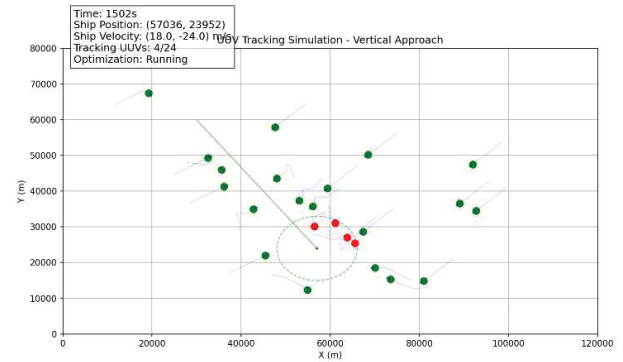


Fig 9. Extended tracking: maximizing persistent detection

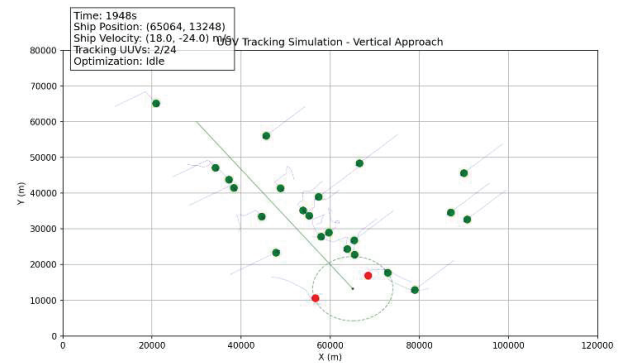


Fig 10. Optimal performance: 50 km continuous tracking range

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